Why the plan exists

Harbor City sits on a low estuary where the river Tamsin meets the sea. The council adopted this plan after the storm of March 2027 flooded 3,900 properties and closed the Dockside rail link for eleven days. Insurers paid out an estimated 186 million dollars for that single event, and two major employers moved their warehouses inland within a year.

The plan sets one headline goal: no more than 200 properties should flood in a storm as severe as 2027 by the year 2040. It combines hard defences, natural storage and managed relocation, because the engineering review concluded that no single measure could reach that goal alone at an affordable price.

Three neighbourhoods carry most of the risk. Dockside holds the port and most of the commercial property. Mill Lane is the lowest residential street in the city, with homes built before any flood standard applied. Saltmeadow is mostly open marsh with a small farming community on its northern edge.

Residents were consulted through nine public meetings and a survey answered by 6,200 households. The survey showed strong support for barrier construction and much weaker support for relocation, which the council has treated as a reason to make any move voluntary and to fund it generously.

The tidal barrier

The centrepiece is the Tamsin Tidal Barrier, a movable gate across the estuary mouth. Construction is due to finish in 2031 at a budgeted cost of 412 million dollars, the largest single item in the plan. The barrier is designed to hold back a surge of 4.6 metres above the mean high tide and to protect about 38,000 homes.

When closed, the barrier stops river water from reaching the sea, so it cannot stay shut for long. Engineers expect it to close around nine times a year at first, rising to about twenty-five closures a year by 2060 as sea levels rise.

Six new pumping stations support the barrier by moving river water past the closed gate. Together they can move 2.4 million litres a minute. The largest, at Pier Road, also serves as a backup control room for the barrier itself.

Each station has its own diesel generator and enough fuel for 72 hours, a requirement added after the 2027 storm cut mains power to the old pumps within the first hour. Annual operating costs for the barrier and pumps together are estimated at 7.5 million dollars.

Natural storage at Saltmeadow

The plan restores 1,150 hectares of marsh at Saltmeadow so that it can flood safely and hold water during a surge. Restoration means removing the old sea wall in stages and replanting native cordgrass. Modelling suggests this lowers peak water levels upstream by around 30 centimetres in a severe storm.

The marsh also absorbs wave energy, which reduces wear on the remaining defences. Farmers on the northern edge will be paid for the land under a twenty-year stewardship scheme instead of being bought out.

Voluntary relocation at Mill Lane

Some homes cannot be protected at a reasonable cost. The plan offers voluntary buyouts to 214 households on Mill Lane at pre-flood market value plus a moving allowance of 15,000 dollars. Families who choose to stay receive grants for raised electrics and flood doors.

Land bought under the scheme will become a public park designed to flood. The council has ruled out compulsory purchase. It will review the scheme every three years and report how many households have taken up the offer.

Paying for the plan

The total capital cost is 690 million dollars. Federal grants cover 55 percent, the state covers 30 percent, and the remaining 15 percent comes from municipal resilience bonds repaid through a small levy on commercial property in protected areas.

The levy was chosen over a general rate rise because commercial property benefits most from the barrier. Households in protected areas pay nothing directly. An independent auditor reports on spending to the council every year.

Monitoring and risks

A network of 38 river and tide sensors feeds a public dashboard, and the plan will be reviewed every five years against updated sea level projections. The main risks named by the review are construction delay, rising steel prices and the possibility that sea level rise outpaces the 2060 design assumption.

If projections worsen, the barrier foundations allow a later height increase of up to one metre. The plan does not address groundwater flooding, which the review describes as a separate problem needing its own study.